12TH BIOLOGY · CHAPTER 15 · SOLVED EXERCISE

Human Nervous System

Complete textbook exercise: MCQs, Short Questions, Long Questions and Inquisitive Questions

SECTION 1: MULTIPLE CHOICE QUESTIONS

1. Neurotransmitter receptors are located on;

(a) Nuclear membrane (b) Nodes of Ranvier
(c) Postsynaptic membrane (d) Myelin sheath

Answer: (c) Postsynaptic membrane — Neurotransmitters cross the synaptic cleft and bind to receptor proteins on the postsynaptic membrane.

2. Which neurotransmitter is released by all motor neurons at their synapses with skeletal muscles?

(a) Acetylcholine (b) GABA
(c) Norepinephrine (d) Dopamine

Answer: (a) Acetylcholine — Acetylcholine is the excitatory neurotransmitter that acts between motor neurons and skeletal muscle fibres.

3. Which part of the nervous system prepares the body during emergency?

(a) Central nervous system (b) Sympathetic nervous system
(c) Parasympathetic nervous system (d) Sensory pathway

Answer: (b) Sympathetic nervous system — The sympathetic system prepares the body for “fight or flight” under physical or emotional stress.

4. Which of these is closest to the brain tissue?

(a) Acetylcholine (b) Arachnoid
(c) Pia mater (d) Cranium

Answer: (c) Pia mater — Pia mater is the innermost layer of the meninges. It adheres to the surface of the brain and spinal cord.

5. In meninges, the cerebrospinal fluid is present between;

(a) Skull and dura mater (b) Dura mater and arachnoid
(c) Arachnoid and pia mater (d) Pia mater and brain or spinal cord

Answer: (c) Arachnoid and pia mater — The cerebrospinal fluid lies below the arachnoid, i.e., between the arachnoid and the pia mater.

6. Which is NOT a function of cerebrospinal fluid?

(a) Cushions brain (b) Produces hormones
(c) Transport nutrients to brain (d) Protects the spinal cord

Answer: (b) Produces hormones — Cerebrospinal fluid cushions and protects the brain and spinal cord and carries nutrients. Hormones are produced by endocrine glands, not by this fluid.

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The textbook does not list the functions of cerebrospinal fluid; this answer is based on general knowledge.

7. During saltatory conduction, the nerve impulse jumps from one _____ to another.

(a) Axon (b) Synapse
(c) Myelin sheath (d) None of Ranvier

Answer: (d) None of Ranvier — In myelinated axons, the action potential jumps from one node of Ranvier to the next.

Option (d) is printed in the textbook as "None of Ranvier". It is a printing mistake for "Node of Ranvier".

8. In a resting neuron, which of these ions has the highest permeability?

(a) Chloride (b) Sodium
(c) Potassium (d) Calcium

Answer: (c) Potassium — In a resting neuron, K^+ ions leak out through the K^+ gates of the membrane, while the Na^+ gates are closed.

9. If nerve impulses in a neuron 'a' are travelling faster than the impulses in neuron 'b', what might be the reason?

(a) Hyperpolarization in neuron 'a'
(b) Neuron 'a' is myelinated
(c) There is a high concentration of sodium ions inside neuron 'a'
(d) Neuron 'a' is thicker than neuron 'b'

Answer: (b) Neuron 'a' is myelinated — A myelinated axon conducts impulses by saltatory conduction, which is up to 50 times faster.

According to the textbook, the velocity is also greater if the diameter of the axon is large, so option (d) is also a possible reason. Myelination has the much larger effect, so (b) is the best answer.

10. Dorsal root of spinal nerve contains?

(a) Sensory neurons (b) Inter-neurons
(c) Motor neurons (d) Sensory and motor neurons

Answer: (a) Sensory neurons — The dorsal root contains the axons and dendrites of sensory neurons. The ventral root contains the axons of motor neurons.

11. Which ions cause the presynaptic vesicles to fuse with the presynaptic membrane and release a neurotransmitter into the synaptic area?

(a) Sodium (b) Potassium
(c) Calcium (d) Iodine

Answer: (c) Calcium — Calcium ions diffuse into the presynaptic axon and make the synaptic vesicles fuse with the membrane.

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12. The most common cause of dementia is;

(a) Alzheimer's disease (b) Brain tumour
(c) Stroke (d) Meningitis

Answer: (a) Alzheimer's disease — Alzheimer's disease is a progressive degenerative brain disorder that leads to dementia in older adults.

13. Nociceptors are specialized receptors that detect:

(a) Light touch (b) Deep pressure
(c) Pain (d) Taste

Answer: (c) Pain — Nociceptors are free nerve endings that detect pain.

14. Which receptor detects deep pressure and vibration?

(a) Meissner's corpuscles (b) Pacinian corpuscles
(c) Merkel cells (d) Ruffini endings

Answer: (b) Pacinian corpuscles — Pacinian corpuscles lie deeper in the skin and detect deep pressure and vibration.

15. Which of the following is a narcotic drug used for pain relief?

(a) Caffeine (b) Morphine
(c) Alcohol (d) Nicotine

Answer: (b) Morphine — Narcotics are opioid drugs used for pain relief, e.g., morphine, codeine and oxycodone.

16. Withdrawal symptoms of drug abuse include:

(a) Improved concentration (b) Headache, nausea, and anxiety
(c) Increased euphoria (d) Faster drug metabolism

Answer: (b) Headache, nausea, and anxiety — Withdrawal symptoms include anxiety, depression, headache, nausea, insomnia and seizures.

17. Caffeine increases alertness by:

(a) Increasing dopamine receptors (b) Blocking adenosine receptors
(c) Stimulating nociceptors (d) Enhancing serotonin degradation

Answer: (b) Blocking adenosine receptors — Caffeine blocks the action of adenosine, the neurotransmitter that promotes sleep and relaxation.

SECTION 2: SHORT QUESTIONS

1. Enlist the basic elements of nervous coordination.

Nervous coordination has three basic elements:

1. **Receptors**, which detect stimuli.
2. **Central nervous system** (brain and spinal cord), which processes the information.
3. **Effectors** (muscles and glands), which produce the response.

2. Define receptors and effectors with examples.

Receptors are the organs, tissues or cells which detect stimuli, i.e., changes in the external and internal environment. Examples: eyes, ears and skin.

Effectors are the parts of the body which produce responses on receiving nerve impulses from the CNS via motor neurons. Examples: muscles (respond by contracting) and glands (respond by producing secretions).

3. State the path of a message transmitted through the nervous system.

Stimulus → Receptor → Sensory neuron → Central nervous system (inter-neurons of brain or spinal cord) → Motor neuron → Effector (muscle or gland) → Response.

4. Define nerve impulse.

A nerve impulse is defined as a wave of electrochemical changes that travels along the length of a neuron.

5. Name the factors responsible for the resting membrane potential of neuron.

1. Fixed anions: large negatively charged molecules (proteins, nucleic acids) inside the cell.
2. Sodium–potassium pump: it pumps three Na^+ out for every two K^+ moved in.
3. Leakage of some K^+ ions out of the cell through the K^+ gates.

6. What are the main components of the nervous system?

The human nervous system has two main components:

- **Central nervous system (CNS):** It consists of the brain and spinal cord.
- **Peripheral nervous system (PNS):** It consists of cranial nerves and spinal nerves, which arise from the brain and spinal cord respectively. The nerves make sensory and motor pathways. The motor pathways are the somatic (voluntary) nervous system and the autonomic nervous system (sympathetic and parasympathetic).

7. Define narcotic drugs and give examples.

A narcotic drug (or simply narcotic) is a type of opioid drug that is used for pain relief and has a high potential for addiction and abuse. Examples: morphine, codeine and oxycodone.

8. Briefly describe the effects of drug addiction and tolerance on the functioning of nervous system.

1. Addictive drugs affect the regions of the brain which regulate pleasure and motivation. When these drugs reach the brain, presynaptic neurons release the neurotransmitter **dopamine**, which produces feelings of pleasure and reward.
2. The drug molecules block the dopamine reabsorption receptors in the presynaptic neuron. So a large amount of dopamine accumulates in the synaptic cleft.
3. The brain adapts and reduces the number of dopamine receptors on the postsynaptic neuron. This leads to **tolerance**, i.e., the user needs higher doses of the drug to get the same pleasure.

4. When the drug is stopped abruptly, dopamine release becomes normal, but due to fewer receptors the postsynaptic neuron is less stimulated. This leads to **withdrawal symptoms**.

9. State the causes, symptoms and treatment of stroke and meningitis.

Stroke

A **stroke** means damage to brain cells due to an interruption in the blood supply to a part of the brain. The interruption may be due to a blockage or a rupture of a blood vessel of the brain.

	Stroke
Causes	High blood pressure, smoking, diabetes, high cholesterol, obesity and family history of stroke.
Symptoms	They vary with the area of brain affected: paralysis on the side of the body opposite to the damaged area; aphasia (inability to express through words); sudden trouble in seeing and walking; dizziness, loss of balance; severe headache.
Treatment	Given as soon as possible: medications (blood thinners, clot-dissolving drugs, antihypertensive drugs); surgery to remove a blood clot or repair a ruptured blood vessel; rehabilitation to recover physical and cognitive function.

Meningitis

Meningitis is an inflammation of the meninges due to infection.

	Meningitis
Causes	Bacterial meningitis (most severe and life-threatening); viral meningitis (usually less severe); fungal meningitis (rare, usually in people with weak immune systems); parasitic infections.
Symptoms	Stiffness in the neck, headache and fever. In severe cases, paralysis or coma.
Treatment	Bacterial: immediate treatment with antibiotics. Viral: usually self-limiting; antiviral medications may be used. Fungal: long-term antifungal medications. Vaccines can prevent some bacterial meningitis.

10. State the causes, symptoms and treatment of brain tumor.

A **brain tumour** is an abnormal growth or mass of cells in the brain. It can be benign (non-cancerous) or malignant (cancerous). It can arise from glial cells or from the neurons themselves.

	Brain tumour
Causes	The exact cause is unknown. Risk factors include exposure to radiation, family history and certain genetic mutations.

	Brain tumour
Symptoms	They depend on the location of the tumour: headaches, severe nerve pain, paralysis, seizures and coma.
Treatment	Surgery, radiation therapy and chemotherapy.

11. State the causes, symptoms and treatment of Alzheimer disease.

Alzheimer's disease is a progressive degenerative brain disorder that affects memory, thinking and behaviour. It leads to dementia in older adults. Its onset usually occurs in aged people, but it is not particularly associated with aging.

	Alzheimer's disease
Causes	A combination of genetic, environmental and lifestyle factors.
Symptoms	It begins with mild memory loss and confusion and progresses to more severe cognitive impairment, including difficulty with language, problem-solving and completing familiar tasks.
Treatment	There is no cure, but there are medicines to slow its progression and to improve memory and cognitive function.

12. Differentiate between;

• Axon and dendrite

Axon	Dendrite
It transmits impulses away from the cell body.	It carries impulses toward the cell body.
It is a long extension of the cell body; a neuron may have a single axon or branching axons.	Dendrites are extensions of the cell body which are often highly branched.
Its end (axon terminal) communicates with a muscle cell, a gland cell or another neuron.	It receives information from other neurons or other cells.

• Sensory neuron, inter-neuron, and motor neurons

Feature	Sensory neuron	Inter-neuron (associative)	Motor neuron
Function	Conducts impulses from receptors to the CNS	Receives impulses from sensory neurons or other inter-neurons and transmits them to motor neurons or other inter-neurons	Conducts impulses away from the CNS to effectors
Location	Cell bodies lie outside the CNS in clusters called ganglia	Occurs entirely within the CNS	Cell bodies lie in the CNS
Structure	Unipolar: one long dendrite on one side of the cell body and an axon on the other	Some have short axons (circuits with nearby neurons); others have long axons (connect different regions of brain and spinal cord)	Multipolar: cell body at one end, many branched dendrites and a long axon

• Myelinated and nonmyelinated neurons

Myelinated neurons	Nonmyelinated neurons
The axon is covered by a myelin sheath, interrupted by nodes of Ranvier.	The axon has no myelin sheath.
Action potentials are produced only at the nodes of Ranvier (saltatory conduction).	Action potentials are reproduced at every adjacent region along the axon.
Impulses travel at speeds of up to 120 metres per second.	Impulses travel at 1 to 3 metres per second.

• White matter and grey matter

White matter	Grey matter
It is composed of myelinated axons.	It contains cell bodies and non-myelinated axons.
In the cerebrum, it lies below the cerebral cortex.	In the cerebrum, it makes the outer cerebral cortex.
In the spinal cord, it makes the outer zone.	In the spinal cord, it makes the inner H-shaped zone.

• Cranial and spinal nerves

Cranial nerves	Spinal nerves
They arise from the brain.	They arise from the spinal cord.
There are 12 pairs.	There are 31 pairs.
Some are sensory, some are motor and others are mixed.	All are mixed nerves.

• Resting membrane potential and active membrane potential

Resting membrane potential	Active membrane potential
It is the membrane potential of a resting (polarized) neuron.	It is the reversed membrane potential of an active neuron (action potential).
The inside of the membrane is negative and the outside is positive.	The inside becomes positive with respect to the outside.
It is about –70 mV.	It reaches about +50 mV.
Na^+ gates are closed; it is maintained by fixed anions, the Na^+–K^+ pump and K^+ leakage.	Na^+ gates open and Na^+ ions rush into the cell.

• Polarization and depolarization

Polarization	Depolarization
It is the state of a resting neuron in which the inner side is negative and the outer side is positive.	It is the reversal of polarity across the membrane, so that the inside becomes positive.
Membrane potential is –70 mV.	Membrane potential rises from –70 mV to zero and then to +50 mV.
No impulse is being conducted.	It produces an action potential (nerve impulse).

• Repolarization and hyperpolarization

Repolarization	Hyperpolarization
Na^+ gates close, K^+ gates open and K^+ ions diffuse out, so the inside becomes negative again.	The membrane potential becomes slightly more negative than the resting potential for a brief period.
It restores the polarity of the membrane after depolarization.	It is due to a slight delay in the closing of all K^+ gates.
It follows depolarization.	It follows repolarization; the neuron cannot generate another action potential (refractory period).

• Presynaptic and postsynaptic neuron

Presynaptic neuron	Postsynaptic neuron
Its axon transmits action potentials towards the synapse.	It lies on the other side of the synapse and receives the message.
Its synaptic knobs contain synaptic vesicles filled with neurotransmitters.	Its membrane has receptor proteins for neurotransmitters.
It releases neurotransmitters into the synaptic cleft.	Neurotransmitters bind to it and generate a new action potential.

• Synaptic knob and synaptic vesicles

Synaptic knob	Synaptic vesicles
It is a swelling at the axon terminal of the presynaptic neuron.	They are small sacs present inside the synaptic knobs.
It contains numerous synaptic vesicles.	They are packed with neurotransmitters.
Its membrane has calcium gates, which open when an action potential arrives.	They fuse with the presynaptic membrane and release neurotransmitters by exocytosis.

• Somatic and autonomic nerves system

Somatic nervous system	Autonomic nervous system
Its motor neurons control skeletal muscles.	Its motor neurons control smooth muscles and glands.
It is voluntary (skeletal muscles can be moved at will).	It is involuntary.
It has no subdivisions.	It has two subdivisions: sympathetic and parasympathetic.

• Drug addiction and drug tolerance

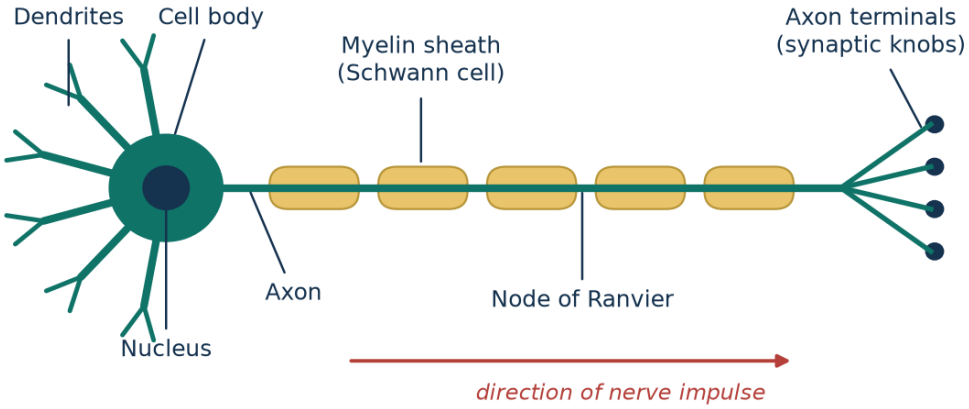
Drug addiction	Drug tolerance
It is the disorder of uncontrollable drug desire and use despite its negative effects.	It is the condition in which the body becomes adaptive and less sensitive to the effects of a drug.
It results from dependence on the drug.	It results from continued use; the brain reduces the number of dopamine receptors.
The person cannot stop using the drug.	The person requires higher doses to achieve the same response.

SECTION 3: LONG QUESTIONS

1. Explain the structure of a neuron.

A neuron (nerve cell) is the structural and functional unit of the nervous system. There are many variations in the structure of neurons, but all neurons have three basic components: a cell body, dendrites and an axon.

- **Cell body:** It is an enlarged region of the neuron. The nucleus and most of the organelles are located in it. Its cytoplasm contains **Nissl's granules**, which are groups of ribosomes and rough endoplasmic reticulum.
- **Dendrites:** These are membrane-bounded extensions of the cell body. They receive information from other neurons or other cells, convert it into nerve impulses and carry it toward the cell body. Motor neurons and inter-neurons receive impulses from many sources at the same time, so they have highly branched dendrites.
- **Axon:** It is a long, membrane-bounded extension of the cell body. It transmits impulses away from the cell body. A neuron may have a single axon or branching axons. The end of an axon is called the **axon terminal**. It may contact and communicate with a muscle cell, a gland cell or another neuron.



Structure of a myelinated neuron (schematic)

Myelin sheath

The special cells which support neurons are called **neuroglia** or glial cells. They supply nutrients to neurons, remove wastes and provide immunity. Two important glial cells are Schwann cells and oligodendrocytes. They wrap around the axons of many neurons and make **myelin sheaths** (sheaths of multiple membrane layers). Schwann cells make myelin sheaths in the peripheral nervous system, and oligodendrocytes make them in the central nervous system.

The myelin sheath is an insulating covering. It helps in the fast transmission of nerve impulses along the axon. It is interrupted at intervals of 1 to 2 mm by small non-myelinated gaps called **nodes of Ranvier**.

2. Explain the function of the three types of neurons with the help of a reflex arc.

Three types of neurons

Feature	Sensory neuron	Inter-neuron (associative)	Motor neuron
Function	Conducts impulses from receptors to the CNS	Receives impulses from sensory neurons or other inter-neurons and transmits them to motor neurons or other inter-neurons	Conducts impulses away from the CNS to effectors
Location	Cell bodies lie outside the CNS in clusters called ganglia	Occurs entirely within the CNS	Cell bodies lie in the CNS
Structure	Unipolar: one long dendrite on one side of the cell body and an axon on the other	Some have short axons (circuits with nearby neurons); others have long axons (connect different regions of brain and spinal cord)	Multipolar: cell body at one end, many branched dendrites and a long axon

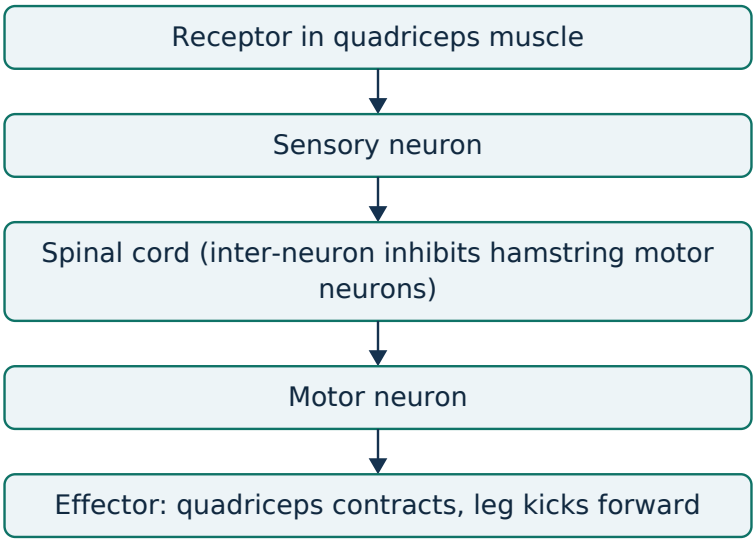
Reflex arc

In the nervous system, neurons follow specific neural pathways (arcs) to receive, transmit and process impulses. The **reflex arc** is the simplest neural pathway. It controls **reflex action**, i.e., an immediate, automatic and involuntary response to an external or internal stimulus.

Example: the knee-jerk reflex

1. When the tendon below the knee (the patellar tendon) is hit with a hammer, the tapping acts as a **stimulus**.
2. The stimulus is sensed by a **receptor** present in the quadriceps muscle of the thigh.
3. The receptor generates a nerve impulse in a **sensory neuron**, which carries the impulse to the spinal cord.
4. In the spinal cord, the sensory neuron transfers the impulse to a **motor neuron** without involving an inter-neuron.
5. The motor neuron carries the impulse to the **effector** (quadriceps muscle), which contracts. The lower leg extends at the knee, i.e., it kicks forward.
6. In the meantime, the sensory neuron also activates an **inter-neuron** in the spinal cord. The inter-neuron releases chemicals which inhibit the motor neurons of the hamstring muscles. So the hamstrings relax during the extension of the lower leg.

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Reflex arc of the knee-jerk reflex

3. Describe the generation and transmission of nerve impulse.

A nerve impulse is a wave of electrochemical changes that travels along the length of a neuron.

1. Before nerve impulse: resting membrane potential

A resting neuron (not conducting a signal) is said to be **polarized**. Its inner side has a negative charge while its outer side has a positive charge. This membrane potential of a polarized neuron is called the **resting membrane potential (RMP)**. It is about –70 mV. The negative sign shows that the inside of the cell is negative with respect to the outside. RMP is established by the following factors:

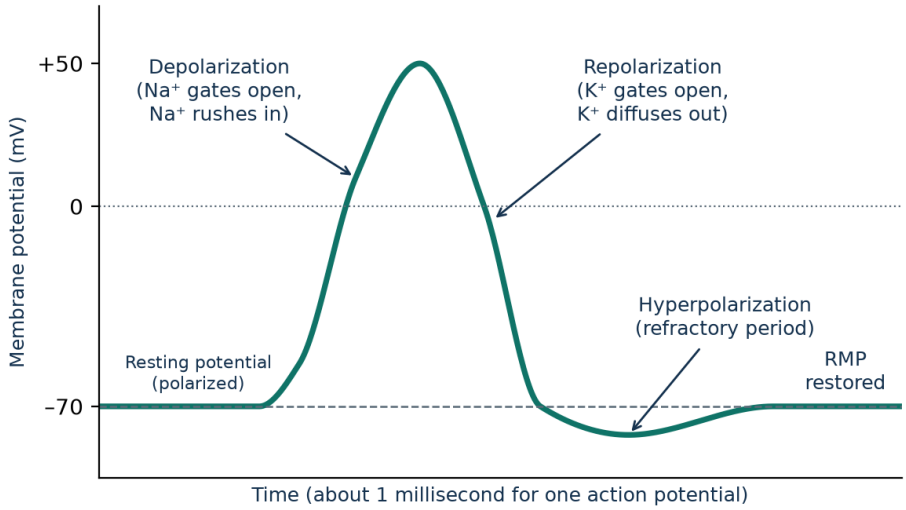
1. **Fixed anions:** Large negatively charged molecules, e.g., proteins and nucleic acids, are more abundant inside the cell and cannot diffuse out.
2. **Sodium–potassium pump:** The concentration of K^+ is 30 times greater inside the neuron than outside, while the concentration of Na^+ is 10 times greater outside than inside. Sodium–potassium pumps constantly move these ions against their concentration gradients by using ATP. For every two K^+ transported inward, three Na^+ are pumped out. So the inside becomes more negative than the outside.
3. **Leakage of K^+:** Some K^+ ions also leak out of the cell through the K^+ gates of the membrane.

2. During nerve impulse: action potential

When a neuron becomes active (generates a nerve impulse), the membrane potential at a region of the neuron is reversed, i.e., the inside becomes positive with respect to the outside. It is called the **active membrane potential** or **action potential**. It is produced by the following events:

- **Depolarization:** The Na^+ gates of the membrane open and Na^+ ions rush into the cell. This reduces the negative charge on the inside. The membrane potential changes from –70 mV towards zero and then reaches +50 mV. This reversal of polarity is called depolarization. It produces an action potential on a short region of the neuron for about one millisecond.

- **Repolarization:** After a short period, the Na^+ gates close and the K^+ gates open. More K^+ ions diffuse out. This makes the inside of the cell negative again (negative inside and positive outside).
- **Hyperpolarization:** Repolarization carries the membrane potential to a value slightly more negative than the resting potential for a brief period. It is due to a slight delay in the closing of all K^+ gates. During hyperpolarization, the neuron cannot generate another action potential. This period is called the **refractory period**.
- **Return to RMP:** After the short refractory period, the original distribution of ions is restored by the sodium–potassium pump and the membrane returns to its RMP, i.e., –70 mV. Now this region is ready for the next action potential.



Graph of an action potential

3. Transmission (propagation) of nerve impulse

In reality, action potentials (impulses) do not travel along the axon. Rather, they are reproduced at adjacent regions along the axon. An action potential in one region of the axon serves as a depolarization stimulus for the next region. So the next region produces its own action potential. Meanwhile, the previous region repolarizes back to the RMP.

The strength (amplitude) of action potentials remains constant along the length of the axon. However, the velocity of the nerve impulse may vary. The velocity is greater if the diameter of the axon is large or if the axon is myelinated.

4. Describe polarization, depolarization and hyperpolarisation of neuron membrane.

- **Polarization:** A resting neuron is polarized. Its inner side has a negative charge and its outer side has a positive charge. The membrane potential is about –70 mV (resting membrane potential). It is established by fixed anions inside the cell, by the sodium–potassium pump (three Na^+ out for every two K^+ in) and by the leakage of some K^+ out of the cell.
- **Depolarization:** When the neuron becomes active, the Na^+ gates open and Na^+ ions rush into the cell. The membrane potential changes from –70 mV towards zero and then reaches +50 mV. This reversal of polarity produces an action potential for

about one millisecond.

- **Repolarization:** The Na^+ gates close and the K^+ gates open. K^+ ions diffuse out and the inside becomes negative again.
- **Hyperpolarization:** Due to a slight delay in the closing of all K^+ gates, the membrane potential becomes slightly more negative than the resting potential for a brief period. During this refractory period, the neuron cannot generate another action potential. Then the sodium–potassium pump restores the RMP (–70 mV).

These changes are shown in the graph of an action potential given in Question 3.

5. Describe the saltatory conduction of nerve impulse.

In myelinated axons, action potentials are produced only at the **nodes of Ranvier**. The action potential jumps from one node of Ranvier to the next, skipping the myelinated regions of the membrane. This is called **saltatory conduction** of the nerve impulse.

Saltatory conduction is up to 50 times faster than the conduction of a nerve impulse through the fastest non-myelinated axons. In humans, non-myelinated neurons conduct impulses at a speed of 1 to 3 metres per second, while myelinated neurons conduct at speeds of up to 120 metres per second.

6. Describe the structure of synapse.

The junction between the axon terminal of one neuron and the dendrites of another neuron, or an effector cell (of a muscle or gland), is called a **synapse**. Its parts are:

- **Presynaptic neuron:** The neuron whose axon transmits action potentials towards the synapse. Its axon terminal has several **synaptic knobs**. These knobs contain numerous **synaptic vesicles**, which are packed with chemicals called neurotransmitters.
- **Synaptic cleft:** At a synapse, the two cells are not in direct contact. There is a fluid-filled gap between them, called the synaptic cleft.
- **Postsynaptic cell:** The neuron or cell on the other side of the synapse. Its membrane has receptor proteins for neurotransmitters.

The synapse between a motor neuron and a muscle fibre is also called a **neuromuscular junction**.

7. Explain synaptic transmission of nerve impulse.

The movement of an action potential across a synapse is called **synaptic transmission**. It happens in the following steps:

1. When an action potential arrives at the synaptic knobs, calcium gates in the presynaptic membrane open.
2. Calcium ions present in the synaptic cleft diffuse rapidly into the presynaptic axon.
3. Due to the entry of calcium ions, the synaptic vesicles fuse with the presynaptic membrane and release neurotransmitters by exocytosis.
4. The released neurotransmitters diffuse rapidly across the cleft and bind to receptor proteins on the postsynaptic membrane.

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5. This binding acts as a message for the postsynaptic membrane. Its Na^+ gates open and an action potential is generated in the postsynaptic membrane.
6. After their action, the neurotransmitters do not remain in the cleft. At many synapses, they are reabsorbed by the presynaptic neuron. At other synapses, they are broken down by enzymes.

8. Classify neurotransmitters as inhibitory and excitatory and list some common examples.

Neurotransmitters are chemical messengers. Neurons use them to communicate with each other and with their target cells. On the basis of their action, they are of two types.

1. Excitatory neurotransmitters

Excitatory neurotransmitters generate a nerve impulse (action potential) in postsynaptic neurons by increasing the permeability of the membrane to Na^+. Examples:

- **Acetylcholine:** acts between motor neurons and skeletal muscle fibres, and between many neurons.
- **Glutamate:** acts in the CNS.
- **Biogenic amines** (they also function as hormones): **norepinephrine** (made in the CNS and sympathetic nerves; regulates sleep patterns, focus and alertness), **epinephrine** (released in adrenal glands and brainstem; plays a role in sleep and alertness), **dopamine** (in the brain; controls body movements, pleasure related to motivation and emotional stimulation) and **serotonin** (about 90% produced in the intestine and 10% in the CNS; regulates appetite, sleep, memory and learning, temperature, mood and behaviour).

2. Inhibitory neurotransmitters

Inhibitory neurotransmitters decrease the chance of transmission of a nerve impulse to the postsynaptic neuron. When they bind to the receptors on the postsynaptic neuron, channels for chloride ions open instead of sodium channels. Chloride ions diffuse into the neuron and make the inside of the membrane even more negative than the RMP. This hyperpolarization causes the inhibitory effect. Examples:

- **GABA (gamma-aminobutyric acid):** the chief inhibitory neurotransmitter of the CNS. It reduces neural excitability, reduces anxiety, stress and fear, and promotes relaxation and sleep.
- **Glycine:** an inhibitory neurotransmitter, especially in the spinal cord, brainstem and retina.
- **Endorphins:** released by neurons in the brain. They reduce the release of the excitatory neurotransmitters that transmit pain messages, so they block pain signals.

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9. Explain briefly the functions of major parts of brain.

Part of brain	Functions
Cerebrum	Frontal lobes: self-control, planning, reasoning, abstract thought. Parietal lobes: process sensory information, body position, attention, motor movements. Temporal lobes: visual memory, language comprehension, emotions. Occipital lobes: vision. The cerebral cortex has motor, sensory and association areas (higher mental activities).
Midbrain	Relays visual and auditory information; its reticular formation connects the hindbrain with the forebrain.
Pons	Transmits impulses between the cerebral hemispheres and the cerebellum.
Medulla oblongata	Relay centre and control centre for heart rate, respiration rate and other homeostatic activities.
Thalamus	Relay centre between sensory information and the cerebrum.
Hypothalamus	Regulates body temperature, hunger and satiety, thirst and emotional states; controls the pituitary gland.
Limbic system	Amygdalae control emotions (love, hate, anger, fear, rage); hippocampus converts short-term memory into long-term memory.
Cerebellum	Processes information about body position, controls posture and coordinates muscle actions, especially rapid and ongoing movements.

10. Describe the structure of human brain and compare it with that of spinal cord.

Structure of the brain

The human brain is enclosed in the cranium and covered by the meninges. It consists of the cerebrum, the brainstem, the diencephalon and the cerebellum.

1. Cerebrum

It is the largest portion of the human brain. Its outer layer is highly folded. A deep groove (fissure) divides it into the left and right **cerebral hemispheres**, which are connected at the base of the groove by the **corpus callosum** (a band of axons). Each hemisphere receives sensory information from the opposite side of the body and sends motor commands to that side. Each hemisphere has four lobes:

- **Frontal lobes:** executive functions including self-control, planning, reasoning and abstract thought.
- **Parietal lobes:** process sensory information, give awareness of the position of the body and the environment, control attention, and plan and execute motor movements.
- **Temporal lobes:** process sensory input for retention of visual memory, language comprehension and emotions.
- **Occipital lobes:** control vision.

The **cerebral cortex** is the folded outer layer of the cerebrum, made of grey matter. It contains **motor areas** (send impulses to voluntary skeletal muscles), **sensory areas** (receive information from different parts, e.g., the auditory cortex in the temporal lobe and the visual cortex in the occipital lobe) and **association areas** (sites of higher mental activities). Below the cortex is white matter, which links regions of the cortex with each other and with other neural centres.

2. Brainstem

The cerebrum is connected to the spinal cord through the brainstem. It lies below the cerebrum and has three divisions:

- **Midbrain:** relays visual and auditory information. It also contains the reticular formation, which connects the hindbrain with the forebrain.
- **Pons:** transmits impulses between the cerebral hemispheres and the cerebellum.
- **Medulla oblongata:** serves as both a relay centre and a control centre for heart rate, respiration rate and other homeostatic activities.

3. Diencephalon

It is present deep within the centre of the brain, above the brainstem and beneath the cerebrum. It has two major parts:

- **Thalamus:** a relay centre between sensory information and the cerebrum.
- **Hypothalamus:** integrates the visceral activities. It regulates body temperature, hunger and satiety, thirst and various emotional states. It also controls the pituitary gland, which in turn regulates many other endocrine glands.

The **limbic system** is a group of linked structures on both sides of the thalamus, just under the cerebrum. It includes the hypothalamus, the amygdala and the hippocampus. The **amygdalae** control feelings and emotions of love, hate, anger, fear, rage and sexual arousal. The **hippocampus** plays a role in the consolidation of information from short-term memory to long-term memory.

4. Cerebellum

It is the second major structure of the brain. It lies below and behind the cerebral hemispheres. It consists of a tightly folded layer of cortex with white matter underneath. It receives sensory impulses from muscles, tendons, joints, eyes and ears, and information from other brain centres. It processes information about body position and controls posture. It coordinates muscle actions, especially rapid and ongoing movements, by acting with the brainstem and the cerebral cortex.

Ventricles

The brain and spinal cord are hollow structures. They contain fluid-filled cavities called **ventricles**. There are four ventricles in the human brain: two large lateral ventricles in the cerebrum, the third ventricle in the diencephalon between the right and left thalamus, and the fourth ventricle between the cerebellum and the brainstem. The fourth ventricle is continuous with the central canal of the spinal cord. The main function of the ventricles is the production and circulation of cerebrospinal fluid.

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Structure of the spinal cord

The spinal cord is a cable of neurons extending from the medulla oblongata of the brain down through the backbone. It is enclosed and protected by the vertebral column and the meninges. There are two zones inside it:

- **Inner zone (grey matter):** It is H-shaped or butterfly-shaped. It contains the cell bodies of inter-neurons and motor neurons, neuroglia and non-myelinated axons. It also contains a central fluid-filled canal, which is a continuation of the fourth ventricle of the brain.
- **Outer zone (white matter):** It contains myelinated axons.

The spinal cord acts as an information highway on which messages are transmitted between the body and the brain.

Comparison of brain and spinal cord

Feature	Brain	Spinal cord
Protection	Enclosed within the cranium and covered by meninges	Enclosed within the vertebral column and covered by meninges
Grey matter	On the outer side (e.g., cerebral cortex)	In the inner zone (H-shaped or butterfly-shaped)
White matter	On the inner side, below the cortex	In the outer zone
Cavity	Four ventricles filled with cerebrospinal fluid	Central canal, continuous with the fourth ventricle
Nerves	12 pairs of cranial nerves	31 pairs of spinal nerves

11. Explain the types and functions of autonomic nervous system.

The autonomic nervous system consists of those motor neurons of spinal and cranial nerves which control smooth muscles and glands. It is involuntary. It controls internal body conditions, e.g., respiration, heartbeat and digestion. It has two subdivisions.

(a) Sympathetic nervous system

It prepares the body under physical or emotional stress. Its motor neurons originate from the thoracic and lumbar regions of the spinal cord and reach different visceral organs (smooth muscles and glands). When it is activated, it prepares the body for **fight or flight**. For example, the heart beats faster and stronger, blood glucose concentration increases, blood flow is diverted to the muscles and heart, and the bronchioles dilate.

(b) Parasympathetic nervous system

It antagonises the response of the sympathetic system after the threat or stress has passed. Its motor neurons originate from the brain and the sacral region of the spinal cord. Many of them are in the vagus (tenth cranial) nerve. They bring the internal organs back to their normal conditions. For example, the heartbeat slows down, blood flow to the heart and skeletal muscles decreases, and the digestive organs increase their secretion and other activities.

Body part	Effect of sympathetic system	Effect of parasympathetic system
Pupil of eye	Dilation	Constriction
Salivary glands	Decreased secretion	Increased secretion
Gastric glands	Inhibition of secretion	Stimulation of secretion
Liver	Stimulation of glucose secretion	Inhibition of glucose secretion
Digestive tract	Decreased motility	Increased motility
Urinary bladder	Relaxation	Contraction
Heart muscle	Increased rate and strength	Decreased rate
Bronchioles	Dilation	Constriction
Blood vessels in muscles	Dilation	No effect
Blood vessels in skin	Constriction	No effect

12. Write a detailed note on the receptors for smell.

Structure

The chemoreceptors for smell (olfaction) are located in the upper part of the nasal cavities. These receptors are actually the ends of sensory neurons. The sensory neurons are present in a thin layer of tissue called the **nasal epithelium**. They are packed tightly between two other types of cells: **supporting cells**, which provide protection, and **basal cells**, which act as stem cells to replace old neurons. The dendrites of these sensory neurons have tiny hair-like structures called **cilia**, which project into the moist mucus lining the nasal cavity.

Functioning

When air enters the nasal cavities, the air-borne chemicals dissolve in the mucus. The dissolved chemicals bind to specific receptor sites on the cilia. This binding creates a nerve impulse in the sensory neurons. The impulse is transmitted to the **olfactory bulb** through the olfactory nerves (cranial nerve I). The olfactory bulb produces the perception of smell.

13. Explain the structure of the receptors for taste.

Structure

The sense of taste is detected by special chemoreceptors called **taste buds**. There are about 10,000 taste buds. Most of them are embedded between bumps called **papillae** on the tongue. Some are present in the throat and on the roof of the oral cavity. A taste bud is bulb-shaped. It is made up of receptor cells with hair-like structures and basal cells (for making new receptor cells). It opens into the mouth through a **taste pore**.

Functioning

When the chemoreceptor cells of taste buds are stimulated by the chemicals of food or drink, they produce nerve impulses in the associated sensory neurons. The impulses travel along cranial nerves VII, IX and X to the brainstem, and from there to the taste-sensing areas of the cerebral cortex. The cortex processes this information with other sensory information, such as smell and texture, to create the complete perception of taste.

Humans have four kinds of taste buds: salty, sweet, sour and bitter. They are concentrated in specific regions of the tongue: sweet at the tip, sour at the sides, bitter at the back, and salty over most of the surface. The four regions overlap at certain places.

14. Describe the receptors for touch and pain.

Receptors for pain (nociceptors)

Nociceptors detect pain. Pain is defined as a stimulus that causes or is about to cause tissue damage. Nociceptors consist of free nerve endings located throughout the body, especially near surfaces. Different nociceptors may respond to extremes of temperature, very intense pressure or specific chemicals (including some released by injured cells).

Receptors for touch

The skin contains a variety of touch receptors. They are **mechanoreceptors** that respond to different physical stimuli.

Touch receptor	Structure and location	Detects
Meissner's corpuscles	Endings of sensory neurons surrounded by a capsule; in the upper layer of skin (lips, fingertips, palm, soles)	Light touch and vibration
Hair follicle receptors	Endings of neurons around the base of hair follicles	Movement of hairs (mechanical or thermal stimuli)
Merkel cells	Oval-shaped receptors in the deepest layer of epidermis; in highly touch-sensitive areas like fingertips	Light touch
Pacinian corpuscles	Sensory neuron endings surrounded by a capsule; deeper in the skin	Deep pressure and vibration
Ruffini endings	Sensory neuron endings enclosed in a capsule; in the dermis	Sustained pressure and stretching

These receptors generate impulses and transmit them via sensory neurons to the spinal cord and then to the brain, where the signals are processed to create the sensation of touch.

15. Describe the use and abuse of narcotic drugs with examples.

A **narcotic drug** is a type of opioid drug that is used for pain relief and has a high potential for addiction and abuse. Narcotics include morphine, codeine and oxycodone.

Use: The term "use" refers to the appropriate or prescribed use of drugs for medical purposes. Narcotics are used for the relief of pain.

Abuse: The term "abuse" generally refers to the inappropriate use of psychoactive drugs. Abuse leads to dependence, addiction and tolerance. The abuse of common psychoactive drugs is given below.

Drug	Source / form	Effect on user	Negative effects
Heroin	Highly addictive opioid from the opium poppy plant; injected, smoked or snorted	False sense of euphoria and relaxation; quickly leads to dependence and addiction	Respiratory depression, infections, mental health disorders
Cannabis	From dried flowering tops, leaves and stem of Indian hemp (*Cannabis sativa*); marijuana, hashish	Relaxation, euphoria, altered perception, increased appetite	Impaired cognitive function, respiratory problems, mental health disorders
Nicotine	Tobacco products: cigarettes, cigars, smokeless tobacco	Addiction and dependence	Respiratory problems, cancers, cardiovascular disease, mental health disorders
Alcohol	Beer, wine and spirits	Relaxation and euphoria	Impaired cognitive and motor functions, liver diseases, cardiovascular diseases, mental health disorders
Inhalants	Nail polish remover, glue (inhaled)	Euphoria, relaxation, altered perception	Respiratory problems, cognitive impairment, neurological damage

Withdrawal symptoms of drug abuse: anxiety, depression, fatigue, agitation, mood swings, tremors, headache, sweating, nausea, loss of appetite, insomnia, rapid heart rate, high blood pressure, hallucinations and seizures.

16. What is meant by drug addiction and drug tolerance? Explain with reference to caffeine and nicotine.

Drug addiction is the disorder of uncontrollable drug desire and use despite its negative effects. The abuse of psychoactive drugs leads to dependence (the person relies on the drug physically or emotionally in order to function), and dependence often results in addiction.

Drug tolerance is the condition in which, as a person continues to use a drug, the body becomes adaptive and less sensitive to its effects. After tolerance, the person requires higher doses to get the same level of response. This leads to increased addiction.

Caffeine

Caffeine is found in many common beverages, such as coffee, tea and energy drinks. It blocks the action of a neurotransmitter called **adenosine**, which promotes sleep and relaxation. As a result, caffeine increases alertness and energy levels. Regular use of caffeine leads to **tolerance**, so the user requires higher doses to achieve the same level of alertness. If caffeine use is stopped abruptly, withdrawal symptoms such as headaches and fatigue occur.

Nicotine

Nicotine is a highly addictive stimulant present in tobacco products (cigarettes, cigars and smokeless tobacco). It binds to nicotine receptors in the brain. These receptors release dopamine and other neurotransmitters, which produce pleasurable feelings. Regular use leads to **tolerance**, so the user requires higher doses to achieve the same pleasure. When nicotine is stopped abruptly, the user experiences withdrawal symptoms such as irritability, anxiety and cravings. Long-term use of nicotine is also associated with an increased risk of cancer, heart disease and respiratory problems.

17. Explain the classification of nervous disorders with examples.

Nervous disorders are classified into five groups:

Type	Description	Examples
Vascular disorders	The blood vessels of the brain and spinal cord are affected.	Stroke, cerebral venous thrombosis
Infectious disorders	Infections of the CNS caused by viruses, bacteria, fungi, protozoa or flatworms.	Meningitis
Structural disorders	Result from abnormalities or damage to the physical structure of the nervous system.	Brain tumour, spinal cord injuries
Functional disorders	Affect the normal functioning of the nervous system without any apparent structural or organic cause.	Headache (e.g., migraine, tension headache)
Degenerative disorders	Progressive deterioration of nerve cells in the brain, spinal cord or peripheral nerves.	Alzheimer’s disease, Parkinson’s disease, Huntington’s disease, multiple sclerosis

- **Stroke:** damage to brain cells due to an interruption in the blood supply to a part of the brain (blockage or rupture of a blood vessel).
- **Meningitis:** inflammation of the meninges due to infection; bacterial meningitis is the most severe.

- **Brain tumour:** an abnormal growth or mass of cells in the brain; it may be benign or malignant.
- **Headache:** pain or discomfort in the head or neck. Primary headaches are due to the headache itself; secondary headaches are due to an underlying structural problem.
- **Alzheimer's disease:** a progressive degenerative brain disorder that affects memory, thinking and behaviour and leads to dementia.

18. Explain the principles and use of EEG, CT scan and MRI.

1. Electroencephalography (EEG)

Principle: EEG measures the electrical activity of the brain. It is based on detecting the electrical nerve impulses being transmitted among the neurons.

Procedure: Metal electrodes are attached to the scalp and connected to an amplifier. The amplifier magnifies the electrical signals detected by the electrodes. The signals are recorded and displayed as a series of waveforms on a computer screen or paper.

Uses: EEG is done to evaluate neurological conditions, including seizures, epilepsy and brain tumours. It is painless and does not require sedation or anaesthesia.

2. Computed tomography (CT) scan

Principle: CT scan uses X-rays and computer processing to generate detailed images of the body's internal structures. It is based on the fact that different tissues of the body absorb X-rays to different degrees.

Procedure: The patient lies on a motorized table that slides into a doughnut-shaped machine called a gantry. The gantry contains an X-ray tube that rotates around the patient. Multiple detectors on the opposite side record the absorption of X-rays through different tissues. A computer uses this information to construct a detailed, cross-sectional image.

Uses: CT scans are useful for imaging soft tissues such as the brain, lungs and abdomen, and for detecting abnormalities in bone structure. They are used to diagnose cancers, infections and injuries.

3. Magnetic resonance imaging (MRI)

Principle: MRI uses a strong magnetic field and radio waves to produce detailed images of the body's internal structures. It is based on the fact that different tissues have different amounts of hydrogen atoms, which have magnetic properties.

Procedure: The patient lies on a motorized table that slides into a large tube-like machine called a scanner. Its powerful magnet creates a strong magnetic field around the body. Radio waves then cause the hydrogen atoms to produce signals, which are detected by a receiver. A computer uses this information to construct a detailed, three-dimensional image.

Uses: MRI is used for imaging soft tissues such as the brain, spinal cord and internal organs, and for detecting abnormalities in blood vessels and certain types of tumours.

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INQUISITIVE QUESTIONS

1. Predict from every day experience what various kinds of receptor can be found in human body.

In daily life we sense many different stimuli, so the body must have a separate kind of receptor for each of them:

Everyday experience	Kind of receptor
Tasting food; smelling a flower	Chemoreceptors (taste buds, olfactory receptors)
Feeling a handshake, pressure or vibration	Mechanoreceptors (touch receptors of skin)
Feeling hot tea or cold water	Thermoreceptors
Feeling the prick of a needle or a burn	Nociceptors (pain receptors)
Seeing light and colours	Photoreceptors (in eyes)
Hearing sounds; keeping balance	Mechanoreceptors (in ears)

Photoreceptors and the receptors of the ear are not described in this chapter; they are added from general knowledge.

2. Conceptualize the electrical activity of brain, which can be recorded using magnets and tomography.

Neurons of the brain communicate by nerve impulses, which are electrical in nature. So an active brain always shows electrical activity. This activity is recorded directly by **EEG**: electrodes on the scalp detect the impulses and show them as waveforms. Active regions of the brain can also be located by imaging. **MRI** uses a strong magnetic field and radio waves, and **CT scan** (computed tomography) uses X-rays, to produce detailed images of the brain. With these tests, doctors can study the working of the brain and diagnose seizures, epilepsy, tumours and injuries without surgery.

3. Justify the way nervous system helps to coordinate complex and intricate movements of hand to play a piano, or write alphabets.

Such movements need the combined action of many parts of the nervous system:

- Receptors in the eyes, ears, skin, muscles and joints send information about the keys or the pen and about the position of the fingers to the CNS through sensory neurons.
- The association areas of the cerebral cortex plan the movement, and the **motor areas** send impulses to the voluntary skeletal muscles of the hand through somatic motor neurons.
- The **cerebellum** receives impulses from muscles, tendons, joints, eyes and ears. It coordinates the muscle actions, especially rapid and ongoing movements, so the fingers move smoothly and in the right order.
- Myelinated neurons carry these impulses very fast (up to 120 m/s), so many muscles can be controlled almost at the same moment.

4. Justify that the development of a modern computer is in fact a product of the understanding of the way nervous coordination occurs in complex organisms like humans.

Nervous system	Computer
Receptors receive stimuli (input).	Input devices (keyboard, mouse, sensors) receive data.
Sensory and motor neurons carry impulses.	Wires and circuits carry electrical signals.
Brain (CNS) processes information and stores memory.	Processor (CPU) processes data; memory stores it.
Effectors (muscles, glands) give the response (output).	Output devices (screen, printer, speakers) give the result.
A neuron either fires an action potential or does not.	A circuit is either on or off (1 or 0).

So a computer follows the same plan as nervous coordination: input → processing → output. Modern artificial "neural networks" are also modelled on the networks of neurons in the brain.

This comparison is not given in the textbook; it is a reasoned answer.

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